

Kuan-Chen Chen

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EDUCATION

University of Florida

Master of Science in Applied Data Science

National Sun Yat-Sen University

Bachelor of Science in Applied Mathematics

Honors & Awards: The Valedictorian of Department of Applied Mathematics, Distinguished Service Award

Gainesville, FL

8/2024 - 5/2026

Kaohsiung, Taiwan

9/2019 - 6/2023

EXPERIENCE

Bioinformatics Lab, University of Florida

Student Research Assistant

- Designed stacking ensemble combining XGBoost, LightGBM, and Balanced Random Forest with logistic regression meta-learner to classify 10,486 CDR3 sequences on highly imbalanced data (1:23), achieving ROC-AUC 0.876
- Engineered 42 physicochemical features from protein sequences using Python and Pandas; optimized classification thresholds under class imbalance, improving precision to 0.716 for high-confidence predictions
- Developed biologically-informed BLOSUM62 augmentation algorithm expanding the minority class $10\times$ ($432 \rightarrow 4,622$ samples), mitigating severe imbalance ($1:23 \rightarrow 1:1.5$) while preventing data leakage in cross-validation

PROJECT

Athlete Career Path Prediction

- Built an interactive Dash dashboard in Python with Pandas for data processing, enabling athletes to compare predicted career trajectories (NBA, NFL, FIFA) using physical attributes across 3 Kaggle datasets
- Engineered cross-sport feature pipeline (BMI, position encoding, age-entered-pro) with scikit-learn and TensorFlow regression models to forecast 10-year player rating trajectories
- Performed exploratory data analysis on salary and performance data from NFL, NBA, and FIFA datasets; standardized measurements and visualized distributions with Matplotlib and Seaborn

cAuldron (Recipe Generation LLM)

- Built an AI recipe app that turns ingredient photos into complete recipes with AI-generated images; shipped two production versions in Python, cutting pipeline latency from 40–80s to 10–20s
- Developed v1 pipeline using CLIP and DETR for multi-ingredient detection across 525+ food classes (88% F1); fine-tuned Llama 3.2 1B via QLoRA on 7,913 curated RecipeNLG recipes, updating only 0.23% of parameters (2.8M) while achieving 92/100 recipe quality score
- Rebuilt v2 on LangGraph with Groq-hosted Llama 4 Scout (vision) and Llama 3.3 70B (generation); parallel fork/join ChromaDB RAG + Tavily search, removing the 4–6 GB VRAM dependency

Natural Leaf Surface Texture Analysis

- Built a dual-application Python pipeline extracting 1,351-dimensional feature vectors (GLCM, fractal dimension, vein geometry, MobileNetV2 embeddings) from 11,000 leaf images for both disease classification (~92% validation accuracy) and engineering surface analysis
- Performed quantitative texture data analysis translating biological surface variability into engineering metrics (roughness proxy, anisotropy index, fractal dimension); applied PCA to 50 components (>95% variance retained) and K-means/hierarchical clustering to identify 5 surface texture groups
- Visualized cluster profiles and texture space with Matplotlib and Seaborn, mapping natural surface patterns to engineered surface analogues (polished, sandblasted, honed, dimpled, brushed finishes)

SKILLS

Languages & ML: Python, SQL, R | PyTorch, scikit-learn, XGBoost, LightGBM, SHAP

LLM & GenAI: HuggingFace Transformers, LoRA/QLoRA, LangGraph, LangSmith, ChromaDB

Web & Viz: Flask, React.js, Gradio, Dash, Shiny, Power BI, Tableau | DB: MySQL, MongoDB